Self-Level Paper 1 Commitments: How Each Governance Commitment Applies at Self Scope in the CKS Three-Level Structure

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Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026). It is the one-hundred-and-first note in Phase B2 of Series B, and the fourth of thirteen notes decomposing B1.20 (recursive Paper 1 commitments) by formalizing how each Paper 1 governance commitment applies specifically at Self scope. It does not introduce new commitments. Its contribution is to specify, in operational form, what each Paper 1 commitment means when its scope object is the Self — the highest governance scope in the three-level structure.

Abstract

Paper 2's architecture is organized around three levels — cell, aspect, and Self — each level inheriting the full set of Paper 1 architectural commitments. B1.20 establishes that this inheritance is recursive: Paper 1's commitments apply at every level, with each level providing the specific scope objects to which each commitment applies. B2.98 formalized the recursive commitment frame; B2.99 formalized cell-level application; B2.100 formalized aspect-level application. This note, B2.101, formalizes Self-level application — the highest governance scope in the three-level structure. At Self scope, A1.01 means humans govern integration architecture through Self DNA; A1.03 means cross-aspect conflicts are registered as first-class; A1.04 means the instinct/reasoning configuration per B2.23 is the Self-scope application of the AI-as-substrate-mediator commitment; A1.07 means Self operations are retraceable through Self lineage; A1.08 means Self substrate is the source of truth for integration behavior; A1.10 means integration outcomes are deterministic given Self integration architecture; A1.12 means aspects perform integration labor while humans govern Self DNA; and A1.13 means Selves satisfy composition requirements at the highest scope. Self-level commitments govern integration behavior — how aspects combine into a coherent intelligence — rather than task or purpose-level behavior. The instinct/reasoning configuration, formalized in B2.23, is the most structurally distinctive Self-level commitment because it specifies how the instinct/reasoning separation per B1.01 operates at deployment scope. B2.24 provides the verification counterpart to this specification.

1. Why Self-level Paper 1 commitments needs standalone formalization

B1.20 establishes that Paper 1's commitments are recursive: the same commitments apply at cell, aspect, and Self scope, with each level's distinct scope objects determining what the commitment concretely means at that level. B2.98 formalized the recursive commitment integrating frame — the structural rule that establishes recursion as architectural requirement. B2.99 formalized cell-level application; B2.100 formalized aspect-level application.

A gap remains. The recursive frame plus cell-level and aspect-level specifications do not, by themselves, constitute a complete specification of Self-level application. Self scope is architecturally distinct in a way that requires its own formalization: the Self is the highest governance scope in the three-level structure, governing integration behavior across aspects rather than governing coordination within a purpose-defined arrangement of cells or governing task execution within a single cell. At Self scope, certain commitments take on forms that have no equivalent at lower levels — most notably, A1.04 (AI-as-substrate-mediator) whose Self-scope application is the instinct/reasoning configuration per B2.23, and A1.01 (human-governed) whose Self-scope application is governance of integration architecture as the deployment's apex governance.

Without a standalone Self-level formalization, the recursive inheritance claim is incomplete as prior art. B2.101 closes that gap by specifying, for each Paper 1 commitment, precisely what it means when its scope object is the Self.

The prior-art function of this note is specific. Any design that claims novel architecture for how governance commitments apply at deployment integration scope must navigate this formalization. The note establishes, as public record, that Self-scope governance under inherited Paper 1 commitments is a specified architectural derivation, not an unclaimed design space.

2. Architectural specification: each Paper 1 commitment at Self scope

The Self's scope object is its integration architecture — the structure by which multiple aspects combine into a coherent intelligence, including Self DNA per B2.21, the instinct/reasoning configuration per B2.23, cross-aspect conflict governance, and cross-level access configuration. What follows applies each Paper 1 commitment to this scope object.

A1.01 — Human-governed at Self scope. Self integration behavior is human-governed through Self DNA per B2.21. Humans govern how the Self integrates aspects by authoring integration architecture, cross-aspect conflict handling rules, and cross-level access configuration. This is the highest-scope instance of the human-governed commitment in the three-level structure: it encompasses the deployment's integrated intelligence. Humans retain the rights to inspect, modify, and override Self integration architecture at all times.

A1.03 — Conflict-as-first-class at Self scope. Cross-aspect conflicts within Self integration are registered as first-class per A1.03. Where aspects generate conflicting outputs or where integration rules produce ambiguity across aspect domains, those conflicts surface as explicit substrate entries rather than being silently resolved. Self integration architecture per B2.21 includes explicit cross-aspect conflict governance. Self-level A1.03 means the Self has governed handling for conflicts that arise specifically at integration scope — conflicts that cannot be resolved within any single aspect.

A1.04 — AI-as-substrate-mediator at Self scope. The instinct/reasoning configuration per B2.23 is the Self-scope application of A1.04. At lower levels, A1.04 specifies how the LLM (instinct layer) and substrate (reasoning layer) interact within cell task execution or aspect coordination. At Self scope, A1.04 specifies how the LLM instances serving the Self's instinct layer and the substrate serving the Self's reasoning layer interact at integration scope — which is precisely what the instinct/reasoning configuration governs. The Self's instinct/reasoning configuration is the substrate specification for AI-and-governance interaction at deployment scope, making B2.23 the concrete instantiation of A1.04 at Self scope.

A1.05 — Tool-agnosticism at Self scope. Self integration behavior is specified in substrate regardless of which specific LLM instances serve the instinct layer. The instinct/reasoning configuration per B2.23 specifies integration behavior independent of specific LLM vendor or model version. Swapping instinct-layer LLM instances does not change the Self's integration architecture or governed integration behavior.

A1.07 — Path retraceability at Self scope. Self operations are retraceable through Self lineage per B2.43. Self integration history is retraceable from the Self's origination through current state. Integration events — cross-aspect coordination decisions, conflict resolutions, governance interventions at integration scope — carry provenance metadata enabling backward trace. Self-level retraceability is the accountability record for the deployment's integrated intelligence.

A1.08 — Substrate-as-source-of-truth at Self scope. Self substrate — comprising Self DNA per B2.21/B2.23, the Self's aspect collection, and cross-level access configuration per B2.21 — is the authoritative source of truth for Self integration behavior. Authoritative integration state is in substrate, not in LLM-internal state, not in ephemeral inference context, and not in aspect-level substrates taken in isolation. The Self substrate is the integration-scope analog of the cell-level substrate's role in Paper 1.

A1.10 — Determinism at Self scope. Given Self integration architecture and the governed behaviors of constituent aspects, Self integration outcomes are deterministic in the CKS sense: same Self substrate state and same integration rules yield equivalent integration behavior. This is representation determinism, not model-output determinism. The determinism contract at Self scope governs integration outputs, not the stochastic outputs of underlying LLM inference.

A1.12 — Labor allocation at Self scope. Aspects — and through aspects, cells — perform Self integration labor. Humans govern Self DNA while aspects execute integration work per A1.12. The LLM instances in the Self's instinct layer perform instinct-layer operations as labor under Self governance. The labor-authority distinction that A1.12 establishes at cell scope holds at Self scope: humans govern Self integration architecture; aspects and LLM instances execute under that governance.

A1.13 — Composition requirements at Self scope. Selves satisfy composition requirements as the highest-level entities in the three-level structure. A Self that violates composition requirements — by allowing integration to operate outside governed substrate, or by bypassing cross-aspect conflict governance — breaks the composition property at the apex scope. Conversely, Selves impose composition requirements on their aspect members: aspects must satisfy their composition requirements within the Self's integration architecture.

A2.01–A2.04 — Governance affordances at Self scope. All four governance affordances apply at Self scope. Inspect (A2.01): humans inspect Self DNA and aspect collection at any time. Modify (A2.02): humans modify Self integration architecture. Override (A2.03): humans override specific Self integration decisions. Rule authoring (A2.04): humans author Self integration rules, including the instinct/reasoning configuration per B2.23. Rule authoring at Self scope — the authoring of Self DNA — is the primary mechanism through which all Self-level Paper 1 commitments are operationally realized.

A2.40 — Provenance at Self scope. Self integration events record provenance per A2.40. Cross-aspect coordination decisions, integration conflict resolutions, and governance interventions at Self scope carry the six-field provenance metadata required by A2.40.

A2.46 — Authoritative content at Self scope. Self DNA is Category 4 authoritative content specifying what integration rules govern this Self. The instinct/reasoning configuration is a named part of Self DNA and is therefore authoritative substrate content, not advisory configuration.

3. What makes Self-level commitments architecturally distinctive

Three features distinguish Self-level Paper 1 commitments from aspect-level and cell-level applications.

Integration scope, not task or purpose scope. At cell scope, the commitments govern a specific informational task. At aspect scope, they govern a purpose-defined arrangement of cells. At Self scope, they govern integration behavior — how multiple aspects combine into a coherent intelligence. Integration scope is the highest scope in the structure, and governance at this scope is what makes the deployment’s integrated intelligence governed rather than emergent from ungoverned combination of individually governed aspects.

A1.04 as instinct/reasoning configuration governance. At cell scope, A1.04 governs how the LLM mediates between human-authored orchestration rules and substrate content within a cell. At Self scope, A1.04 governs the instinct/reasoning configuration — which LLM instances serve as instinct layer for the Self and how the substrate (reasoning layer) mediates instinct at integration scope. This is the most architecturally explicit manifestation of A1.04 in the three-level structure, because the instinct/reasoning separation per B1.01 is itself most visible at deployment scope where the entire Self’s intelligence integrates.

Governance apex. The Self is the highest governance scope in the three-level structure. Self-level governance encompasses the deployment’s integrated intelligence governance. There is no higher-scope governance entity in the three-level structure; the Self-level application of each commitment is the terminal application for a given deployment.

4. The biological analog

The biological analog for Self-level commitments is organism-level regulatory governance. A biological organism does not merely aggregate organs; it maintains organism-level regulatory mechanisms — hormonal systems, neural integration, immune response — that govern how organs combine into coherent life function. These organism-level mechanisms apply the same biological principles as cell-level and tissue-level mechanisms, but at organism scope and with organism-scope objects. The organism's organism-level governance is not a repetition of cell governance; it governs integration across organ systems.

CKS Self-level commitments are the governed architectural analog. The Self does not merely aggregate aspects; it maintains integration architecture — Self DNA per B2.21, instinct/reasoning configuration per B2.23, cross-aspect conflict governance — that governs how aspects combine into coherent intelligence. The same Paper 1 principles apply at Self scope with Self-scope objects, in the same way that biological regulatory principles apply at organism scope with organism-scope mechanisms.

The biological analog is a conceptual scaffold rather than a technical claim. What the architecture commits to is governed integration behavior under inherited Paper 1 commitments. The organism analogy names the structural parallel; the architectural substance is the full commitment specification in §2.

5. Inherited Paper 1 commitments

All Paper 1 commitments apply at Self scope. B1.20 establishes recursive inheritance; B2.98 formalizes the integrating frame that makes that inheritance a structural rule rather than a claim requiring fresh argument at each level. This note provides the substantive content — what each commitment means at Self scope. The note does not enumerate commitments not in Paper 1 or Paper 2 and does not modify inherited commitments; it specifies their Self-scope application.

The verification counterpart to this specification is B2.24 (Self-level inheritance verification). B2.24 provides the verification framework for confirming that a given Self satisfies Paper 1 commitments at its scope. B2.101 specifies what commitments apply; B2.24 verifies they are met. The two notes are complementary and together constitute a complete Self-level treatment: specification without verification is incomplete; verification without specification has no object.

6. Operational implications

Five operational implications follow from Self-level Paper 1 commitments.

Self-level governance review granularity. Governance reviews of a deployment's integration behavior focus on Self-level commitment satisfaction: Is integration architecture held in human-governed substrate? Are cross-aspect conflicts registered as first-class? Is the instinct/reasoning configuration authored under A2.04? Self-level governance reviews address integration-scope behaviors; they do not re-examine cell-level or aspect-level behaviors, which are governed at their own scope.

Self DNA authoring as primary governance mechanism. Rule authoring per A2.04 at Self scope is the primary mechanism for Self-level commitment application. The instinct/reasoning configuration per B2.23 is the named part of Self DNA whose authoring most directly instantiates A1.04 at Self scope. Each governance affordance — inspect, modify, override, rule authoring — is exercisable on Self DNA at any time.

Self-level compliance demonstration. Compliance with Paper 1 commitments is demonstrable at integration granularity. A deployment can demonstrate, from Self DNA and Self lineage, that each commitment is satisfied at Self scope without needing to re-verify cell-level or aspect-level compliance in the same demonstration.

Self lineage for retraceability. Self lineage per B2.43 provides the operational history for Self-level retraceability per A1.07. Integration events are retraceable through Self lineage independent of aspect-level or cell-level lineage, enabling backward trace from current integration behavior to originating governance decisions.

Non-duplication of lower-level governance. Self-level governance reviews do not re-examine cell-level or aspect-level commitment satisfaction. Compliance demonstrations at Self scope address integration-scope behaviors; lower-level compliance is governed at its own level under the recursive-levels principle established in B1.20 and B2.98.

7. Limits

Self-level commitments do not duplicate lower-level commitments. They are the same commitments at a different scope with different scope objects. The scope distinction matters architecturally: A1.04 at Self scope has a concrete meaning — instinct/reasoning configuration governance — that is distinct from A1.04 at aspect scope or at cell scope. Treating Self-level application as mere repetition of cell-level application loses the scope-specific content that distinguishes integration governance from task governance.

Self-level governance does not replace lower-level governance. The three levels each govern their own scope under the recursive-levels principle. Self governance governs integration; aspect governance governs purpose-scope coordination; cell governance governs task-level operations. These are distinct scopes with distinct governed behaviors. Self governance does not subsume lower-level governance; the recursive structure requires all three levels to be governed independently at their own scope.

B2.24 is the verification counterpart. This note specifies what commitments apply at Self scope and what their Self-scope application means. It does not specify how to verify that a given Self satisfies those commitments; that is B2.24's function. Implementations should treat B2.101 (specification) and B2.24 (verification) as complementary rather than substitutable.

No new commitments. This note introduces no architectural commitments beyond what Paper 1 and Paper 2 establish. Every claim above is traceable to a named commitment in Paper 1 (A1.01–A2.46) or to the recursive-levels principle established in Paper 2 (§5.2) and formalized in B1.20 and B2.98.

8. Operational test

A deployment's Self instantiates the Self-level Paper 1 commitments if and only if all of the following are true:

1. Self integration architecture is held in human-governed substrate (Self DNA per B2.21), with humans retaining the rights to inspect, modify, and override integration rules at all times (A1.01 at

Self scope).

2. Cross-aspect conflicts arising in integration are registered as explicit substrate entries rather than silently resolved (A1.03 at Self scope).
3. The instinct/reasoning configuration per B2.23 is authored under human authority and specifies which LLM instances serve the instinct layer and how the reasoning substrate mediates instinct at integration scope (A1.04 at Self scope).
4. Self integration behavior is specified in substrate independent of specific LLM vendor or model version (A1.05 at Self scope).
5. Self integration events carry provenance metadata enabling backward trace through Self lineage per B2.43 (A1.07 at Self scope).
6. The authoritative source of truth for Self integration behavior is Self substrate, not LLM-internal state or ephemeral inference context (A1.08 at Self scope).
7. Given Self integration architecture, integration outcomes are deterministic in the representation sense (A1.10 at Self scope).
8. Integration labor is performed by aspects and LLM instances under human governance of Self DNA, not by humans performing integration labor that bypasses Self DNA (A1.12 at Self scope).
9. The Self satisfies composition requirements at the highest scope and imposes composition requirements on its aspect members (A1.13 at Self scope).

A deployment whose Self fails any of (1)–(9) does not satisfy Paper 1 commitments at Self scope. It may be a functional deployment; it is not a CKS-governed Self.

9. Conclusion

Self-level Paper 1 commitments are not a special category; they are the application of Paper 1's commitments to the highest-scope object in the three-level structure. The Self governs integration behavior — how aspects combine into a coherent intelligence — and every Paper 1 commitment applies to that governance. The architecturally distinctive features of Self-level application are: governance of integration scope rather than task or purpose scope; A1.04's concrete form as instinct/reasoning configuration governance; and the governance-apex property that makes Self governance the terminal governance scope for a deployment's integrated intelligence.

This note is the fourth of thirteen notes decomposing B1.20. B2.102 formalizes recursive governance A1.01 as a standalone operational variant. B2.103–B2.108 provide per-commitment deep-dive formalizations. B2.109 formalizes recursive operational tests. B2.110, the final note of Phase B2, formalizes recursive commitments verification and closes the phase. Together these thirteen notes constitute the complete B1.20 decomposition — from the integrating recursive frame through each level's application through full operational and verification specification.

The prior-art function served by B2.101 is precise: the formalization that Self-scope governance inherits and applies each Paper 1 commitment, with the instinct/reasoning configuration as the structurally distinctive Self-scope A1.04 application, is now public record.

Source papers

Li, W. (2026). Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance. April 2026. ORCID: 0009-0004-8065-3235.

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